SG No. 194 D_{6h}^4 $P6_3/mmc$ [hexagonal]

* plus set: $+[0, 0, 0]$

Table 1: Wyckoff site: `2a`, site symmetry: `-3m.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,7,8,9,13,14,15,19,20,21]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[4,5,6,10,11,12,16,17,18,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 5: Wyckoff site: `4e`, site symmetry: `3m.`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,19,20,21]`
2	$\left[0, 0, z + \frac{1}{2}\right]$	`[4,5,6,22,23,24]`
3	$[0, 0, -z]$	`[7,8,9,13,14,15]`
4	$\left[0, 0, \frac{1}{2} - z\right]$	`[10,11,12,16,17,18]`

Table 6: Wyckoff site: `4f`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3,19,20,21]`
2	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	`[4,5,6,22,23,24]`
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[7,8,9,13,14,15]`
4	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	`[10,11,12,16,17,18]`

Table 7: Wyckoff site: `6g`, site symmetry: `.2/m.`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,8,13,20]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,7,14,19]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,9,15,21]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,11,16,23]`
5	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[5,10,17,22]`
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[6,12,18,24]`

Table 8: Wyckoff site: `6h`, site symmetry: `mm2`

No.	position	mapping
1	$\left[x, 2x, \frac{1}{4}\right]$	`[1,11,16,20]`
2	$\left[-2x, -x, \frac{1}{4}\right]$	`[2,10,17,19]`
3	$\left[x, -x, \frac{1}{4}\right]$	`[3,12,18,21]`
4	$\left[-x, -2x, \frac{3}{4}\right]$	`[4,8,13,23]`
5	$\left[2x, x, \frac{3}{4}\right]$	`[5,7,14,22]`
6	$\left[-x, x, \frac{3}{4}\right]$	`[6,9,15,24]`

Table 9: Wyckoff site: `12i`, site symmetry: `.2.`

No.	position	mapping
1	$[x, 0, 0]$	`[1,8]`
2	$[0, x, 0]$	`[2,7]`
3	$[-x, -x, 0]$	`[3,9]`
4	$\left[-x, 0, \frac{1}{2}\right]$	`[4,11]`
5	$\left[0, -x, \frac{1}{2}\right]$	`[5,10]`
6	$\left[x, x, \frac{1}{2}\right]$	`[6,12]`
7	$[-x, 0, 0]$	`[13,20]`
8	$[0, -x, 0]$	`[14,19]`
9	$[x, x, 0]$	`[15,21]`
10	$\left[x, 0, \frac{1}{2}\right]$	`[16,23]`
11	$\left[0, x, \frac{1}{2}\right]$	`[17,22]`
12	$\left[-x, -x, \frac{1}{2}\right]$	`[18,24]`

Table 10: Wyckoff site: `12j`, site symmetry: `m..`

No.	position	mapping
1	$\left[x, y, \frac{1}{4}\right]$	`[1,16]`
2	$\left[-y, x-y, \frac{1}{4}\right]$	`[2,17]`
3	$\left[-x+y, -x, \frac{1}{4}\right]$	`[3,18]`
4	$\left[-x, -y, \frac{3}{4}\right]$	`[4,13]`
5	$\left[y, -x+y, \frac{3}{4}\right]$	`[5,14]`
6	$\left[x-y, x, \frac{3}{4}\right]$	`[6,15]`
7	$\left[y, x, \frac{3}{4}\right]$	`[7,22]`

continued ...

Table 10

No.	position	mapping
8	$\left[x-y,\ -y,\ \frac{3}{4}\right]$	[8,23]
9	$\left[-x,\ -x+y,\ \frac{3}{4}\right]$	[9,24]
10	$\left[-y,\ -x,\ \frac{1}{4}\right]$	[10,19]
11	$\left[-x+y,\ y,\ \frac{1}{4}\right]$	[11,20]
12	$\left[x,\ x-y,\ \frac{1}{4}\right]$	[12,21]

Table 11: Wyckoff site: 12k, site symmetry: .m.

No.	position	mapping
1	$[x,\ 2x,\ z]$	[1,20]
2	$[-2x,\ -x,\ z]$	[2,19]
3	$[x,\ -x,\ z]$	[3,21]
4	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	[4,23]
5	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	[5,22]
6	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	[6,24]
7	$[2x,\ x,\ -z]$	[7,14]
8	$[-x,\ -2x,\ -z]$	[8,13]
9	$[-x,\ x,\ -z]$	[9,15]
10	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	[10,17]
11	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	[11,16]
12	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	[12,18]

Table 12: Wyckoff site: 24l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x-y,\ z]$	[2]
3	$[-x+y,\ -x,\ z]$	[3]
4	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[4]
5	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	[5]
6	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	[6]
7	$[y,\ x,\ -z]$	[7]
8	$[x-y,\ -y,\ -z]$	[8]
9	$[-x,\ -x+y,\ -z]$	[9]
10	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[10]
11	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	[11]
12	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	[12]
13	$[-x,\ -y,\ -z]$	[13]
14	$[y,\ -x+y,\ -z]$	[14]
15	$[x-y,\ x,\ -z]$	[15]
16	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[16]
17	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	[17]
18	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[18]
19	$[-y,\ -x,\ z]$	[19]

continued ...

Table 12

No.	position	mapping
20	$[-x+y,\, y,\, z]$	[20]
21	$[x,\, x-y,\, z]$	[21]
22	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[22]
23	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	[23]
24	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	[24]